

Problem - 4

Solution

$$\dot{S} = -r[H, S] + S_0 \Gamma_S \Gamma_S^* - \frac{1}{2} \dot{q} \Gamma^* \Gamma, S^2$$

$$= r\sigma - S(r\sigma -)^+ - \frac{1}{2} \dot{q} \Gamma^* \Gamma S^2$$

$$= r\sigma - S r\sigma + - \frac{1}{2} \dot{q} (\Gamma^* \Gamma S^2)$$

$$= (r\sigma)^2 \left[\sigma - S\sigma + - \frac{1}{2} \dot{q} \Gamma^* \Gamma S^2 \right]$$

$$= r \left[\sigma - S\sigma + - \frac{1}{2} \dot{q} \Gamma^* \Gamma S^2 \right]$$

$$= r \left[\sigma - S\sigma + - \frac{1}{2} (\Gamma^* \Gamma S + S \Gamma^* \Gamma) \right]$$

$$= r \left[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \frac{1}{2} \left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{pmatrix} + \begin{pmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right) \right]$$

$$= r \left[\begin{pmatrix} S_{11} & 0 \\ 0 & 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 & S_{01} \\ S_{10} & 2S_{11} \end{pmatrix} \right]$$

$$= r \begin{bmatrix} S_{11} & -\frac{S_{01}}{2} \\ -\frac{S_{10}}{2} & -S_{11} \end{bmatrix}$$

$$= \begin{bmatrix} rS_{11} & -\frac{rS_{01}}{2} \\ -\frac{rS_{10}}{2} & -rS_{11} \end{bmatrix}$$

$$\dot{S} = \frac{dS}{dt} = \begin{bmatrix} S_{11}(0) e^{-rt} & S_{01}(0) e^{-\frac{rt}{2}} \\ S_{10}(0) e^{-\frac{rt}{2}} & S_{11}(0) e^{-rt} \end{bmatrix}$$

$$Z(s) = M_0 s M_0 + M_1 s M_1 +$$

$$= \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda-1} \end{pmatrix} \begin{pmatrix} s_{00} & s_{0r} \\ s_{r0} & s_{rr} \end{pmatrix} \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda-1} \end{pmatrix} + \begin{pmatrix} 0 & \overline{\lambda} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} s_{00} & s_{0r} \\ s_{r0} & s_{rr} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \overline{\lambda} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} s_{00} & \overline{\lambda-1} s_{0r} \\ \overline{\lambda-1} s_{r0} & (\lambda-1) s_{rr} \end{pmatrix} + \begin{pmatrix} \lambda s_{rr} & 0 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} s_{00} + \lambda s_{rr} & \overline{\lambda-1} s_{0r} \\ \overline{\lambda-1} s_{r0} & (\lambda-1) s_{rr} \end{pmatrix}$$

where, $\lambda = \lambda - e^{-rt} \Rightarrow \overline{\lambda-1} = e^{-\frac{rt}{2}}$

$$Z(s) = \begin{pmatrix} s_{00} + (\lambda - e^{-rt}) s_{rr} & \overline{\lambda-1} s_{0r} \\ \overline{\lambda-1} s_{r0} & (\lambda-1) s_{rr} \end{pmatrix}$$

$$= \begin{pmatrix} s_{00} + s_{rr} - s_{rr} e^{-rt} & e^{-\frac{rt}{2}} s_{0r} \\ e^{-\frac{rt}{2}} s_{r0} & (\lambda - \lambda + e^{-rt}) s_{rr} \end{pmatrix}$$

$$= \begin{pmatrix} \lambda - s_{rr} e^{-rt} & s_{0r} e^{-\frac{rt}{2}} \\ s_{r0} e^{-\frac{rt}{2}} & s_{rr} e^{-rt} \end{pmatrix}$$

$\frac{d(s)}{dt} = Z(s)$

b) Solution

$$Z(s) = (s-1)S + P \pm 0T$$

$$= (s-1) \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} + P \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left[1 - \frac{(1-e^{-2rt})}{2} \right] \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} + \left(\frac{1-e^{-2rt}}{2} \right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left(1 - \frac{1}{2} + \frac{e^{-2rt}}{2} \right) \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} + \left(\frac{1-e^{-2rt}}{2} \right) \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left(\frac{1+e^{-2rt}}{2} \right) \begin{pmatrix} 800 & 801 \\ 820 & 811 \end{pmatrix} + \left(\frac{1-e^{-2rt}}{2} \right) \begin{pmatrix} 800 & -801 \\ -820 & 811 \end{pmatrix}$$

$$= \frac{1}{2} \left[\begin{pmatrix} 800(1+e^{-2rt}) & 801(1+e^{-2rt}) \\ 820(1+e^{-2rt}) & 811(1+e^{-2rt}) \end{pmatrix} + \begin{pmatrix} 800(1-e^{-2rt}) & 801(e^{-2rt}-1) \\ 820(e^{-2rt}-1) & 811(1-e^{-2rt}) \end{pmatrix} \right]$$

$$= \frac{1}{2} \left[\begin{pmatrix} 800(1+1+e^{-2rt}-e^{-2rt}) & 801(1+e^{-2rt}+e^{-2rt}-1) \\ 820(1+e^{-2rt}+e^{-2rt}-1) & 811(1+1+e^{-2rt}-e^{-2rt}) \end{pmatrix} \right]$$

$$= \frac{1}{2} \begin{bmatrix} 2 \cdot 800 & 2 \cdot 801 e^{-2rt} \\ 2 \cdot 820 e^{-2rt} & 2 \cdot 811 \end{bmatrix}$$

$$= \begin{pmatrix} 800 & 801 e^{-2rt} \\ 820 e^{-2rt} & 811 \end{pmatrix} = \begin{bmatrix} 800 & 801 e^{-2rt} \\ 820 e^{-2rt} & 811 \end{bmatrix}$$

$$\dot{\mathcal{S}} = \frac{d\mathcal{S}}{dt} = -i[H, \mathcal{S}] + \sum_j \Gamma_j \mathcal{S} \Gamma_j^\dagger - \frac{1}{2} \sum_j \Gamma_j^\dagger \Gamma_j \mathcal{S}$$

$$\dot{\mathcal{S}} = \Gamma \mathcal{S} \Gamma^\dagger - \frac{1}{2} \sum_j \Gamma_j^\dagger \Gamma_j \mathcal{S}$$

$$= \Gamma \mathcal{S} \Gamma^\dagger - \frac{1}{2} \sum_j \Gamma_j^\dagger \Gamma_j \mathcal{S}$$

$$= (\Gamma)^\dagger \mathcal{S} \Gamma - \frac{1}{2} \sum_j (\Gamma_j)^\dagger \mathcal{S} \Gamma_j$$

where

$$= \Gamma [\mathcal{S} \Gamma - \frac{1}{2} \sum_j \Gamma_j^\dagger \mathcal{S} \Gamma_j]$$

$$\Gamma_j^\dagger = \Gamma$$

$$= \Gamma [\mathcal{S} \Gamma - \frac{1}{2} \sum_j \Gamma_j^\dagger \mathcal{S} \Gamma_j]$$

$$= \Gamma \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathcal{S}_{00} & \mathcal{S}_{01} \\ \mathcal{S}_{10} & \mathcal{S}_{11} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \frac{1}{2} (\mathbb{I} \mathcal{S} + \mathcal{S} \mathbb{I}) \right]$$

$$= \Gamma \left[\begin{pmatrix} \mathcal{S}_{00} & -\mathcal{S}_{01} \\ -\mathcal{S}_{10} & \mathcal{S}_{11} \end{pmatrix} - \frac{1}{2} (2\mathcal{S}) \right]$$

$$= \Gamma \left[\begin{pmatrix} \mathcal{S}_{00} & -\mathcal{S}_{01} \\ -\mathcal{S}_{10} & \mathcal{S}_{11} \end{pmatrix} - \begin{pmatrix} \mathcal{S}_{00} & \mathcal{S}_{01} \\ \mathcal{S}_{10} & \mathcal{S}_{11} \end{pmatrix} \right]$$

$$= \Gamma \begin{bmatrix} 0 & -2\mathcal{S}_{01} \\ -2\mathcal{S}_{10} & 0 \end{bmatrix}$$

$$\dot{\mathcal{S}} = \begin{bmatrix} 0 & -2r\mathcal{S}_{01} \\ -2r\mathcal{S}_{10} & 0 \end{bmatrix} \quad \mathcal{P} \quad \frac{d\mathcal{S}}{dt} = \begin{pmatrix} \frac{d\mathcal{S}_{00}}{dt} & \frac{d\mathcal{S}_{01}}{dt} \\ \frac{d\mathcal{S}_{10}}{dt} & \frac{d\mathcal{S}_{11}}{dt} \end{pmatrix}$$

$$\dot{\mathcal{S}} = \begin{pmatrix} \mathcal{S}_{00} & \mathcal{S}_{01} e^{-2rt} \\ \mathcal{S}_{10} e^{-2rt} & \mathcal{S}_{11} \end{pmatrix} = \mathcal{S}(t)$$

$$\underline{\underline{\dot{\mathcal{S}} = \mathcal{S}(t)}}$$

2) Generalized measurement

Solution

$$M_+ = \sqrt{1/2} |+\rangle \langle +| = \sqrt{1/2} (|+\rangle \langle +|)$$

$$U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_z(\theta)$$

$$= |0\rangle\langle 0| \otimes [|0\rangle\langle 0| + |1\rangle\langle 1|] + |1\rangle\langle 1| \otimes (e^{-i\theta/2} |0\rangle\langle 0| + e^{i\theta/2} |1\rangle\langle 1|)$$

$$= |00\rangle\langle 00| + |01\rangle\langle 01| + |1\rangle\langle 1| \otimes (e^{-i\theta/2} |0\rangle\langle 0| + e^{i\theta/2} |1\rangle\langle 1|)$$

$$= |00\rangle\langle 00| + |01\rangle\langle 01| + |1\rangle\langle 1| \otimes (e^{-i\theta/2} |0\rangle\langle 0| + e^{i\theta/2} |1\rangle\langle 1|)$$

$$= |00\rangle\langle 00| + |01\rangle\langle 01| + e^{-i\theta/2} |10\rangle\langle 10| + e^{i\theta/2} |11\rangle\langle 11|$$

$$M_+ = \frac{1}{\sqrt{2}} (|+\rangle \langle +|) [|00\rangle\langle 00| + |01\rangle\langle 01| + e^{-i\theta/2} |10\rangle\langle 10| + e^{i\theta/2} |11\rangle\langle 11|] \frac{1}{\sqrt{2}} (|+\rangle \langle +|)$$

$$= |0\rangle\langle 0| + \frac{(e^{-i\theta/2} + e^{i\theta/2})}{2} |1\rangle\langle 1|$$

$$= |0\rangle\langle 0| + \cos\left(\frac{\theta}{2}\right) |1\rangle\langle 1|$$

$$M_- = \sqrt{1/2} |-\rangle \langle -|$$

$$= \frac{1}{\sqrt{2}} (|-\rangle \langle -|) [|00\rangle\langle 00| + |01\rangle\langle 01| + e^{-i\theta/2} |10\rangle\langle 10| + e^{i\theta/2} |11\rangle\langle 11|] \frac{1}{\sqrt{2}} (|-\rangle \langle -|)$$

$$= \frac{1}{2} [\underbrace{|0\rangle\langle 0| - |0\rangle\langle 0|}_{=0} + e^{-i\theta/2} |1\rangle\langle 1| - e^{i\theta/2} |1\rangle\langle 1|]$$

$$= \frac{1}{2} [(e^{-i\theta/2} - e^{i\theta/2})] |1\rangle\langle 1|$$

$$M_- = - \left(\frac{e^{i\frac{gt}{2}} - e^{-i\frac{gt}{2}}}{2i} \right) | \hbar > \hbar \hbar | \quad \text{remember}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$M_- = -i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar |$$

$$M_- = -i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar |$$

Therefore

$$\begin{aligned} E_+ &= M_+^\dagger M_+ = \left(| \hbar > \hbar \hbar | + \cos\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \right)^\dagger \left(| \hbar > \hbar \hbar | + \cos\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \right) \\ &= | \hbar > \hbar \hbar | + \cos^2\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \end{aligned}$$

$$\begin{aligned} E_- &= M_-^\dagger M_- = \left(-i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \right)^\dagger \left(-i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \right) \\ &= -i^\dagger \left(\sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \right) (-i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar |) \\ &= -(-i) \left(\sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | (-i \sin\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar |) \right) \\ &= \sin^2\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \end{aligned}$$

Based on the properties of QM

$$\sum_i E_i = I$$

$$\begin{aligned} \sum E &= | \hbar > \hbar \hbar | + \cos^2\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | + \sin^2\left(\frac{gt}{2}\right) | \hbar > \hbar \hbar | \\ &= | \hbar > \hbar \hbar | + \underbrace{\left[\cos^2\left(\frac{gt}{2}\right) + \sin^2\left(\frac{gt}{2}\right) \right]}_{=1} | \hbar > \hbar \hbar | \end{aligned}$$

$$= | \hbar > \hbar \hbar | + | \hbar > \hbar \hbar | = I \quad \checkmark \quad 6)$$

$$\sum E = |0240\rangle + |1222\rangle$$

$$= I$$

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Conservation of probability operator is $P^2 = P$

$$E_+^2 = E_+$$

$$E_-^2 = E_-$$

let's

$$E_+^2 = E_+$$

$$|0240\rangle + \cos^4\left(\frac{\theta}{2}\right)|1222\rangle = |0240\rangle + \cos^2\left(\frac{\theta}{2}\right)|1222\rangle$$

$$\cos^4\left(\frac{\theta}{2}\right)|1222\rangle = \cos^2\left(\frac{\theta}{2}\right)|1222\rangle = |0240\rangle - |0220\rangle$$

$$\left[\cos^4\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right]|1222\rangle = 0$$

$$\cos^4\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) = 0$$

$$\cos^2\left(\frac{\theta}{2}\right)\left(\cos^2\left(\frac{\theta}{2}\right) - 1\right) = 0$$

Two conditions

$$1 - \cos^2\left(\frac{\theta}{2}\right) = 0$$

$$1 - \cos^2\left(\frac{\theta}{2}\right) - 1 = 0$$

for case-1

$$\cos^2\left(\frac{\theta}{2}\right) = 0$$

$$\left(\frac{\theta}{2}\right) = \cos^2(0)$$

$$\frac{\theta}{2} = \frac{\pi}{2} \Rightarrow \underline{\underline{\theta = \pi}}$$

7)

$$a) \quad \theta_t = \pi$$

For case - 2

$$\cos^2(\theta_t) = 2$$

$$\left(\frac{\theta_t}{2}\right) = (\cos(\theta_t))^2$$

$$\frac{\theta_t}{2} = n\pi$$

$$\underline{\underline{\theta_t = 2n\pi \text{ for } n \in \mathbb{Z}}}$$

b) Solution

$$E_+ = 107201 + \cos^2(\theta_t) 127221$$

$$E_- = \sin^2(\theta_t) 127221$$

$$\theta_t = \pi$$

$$\begin{aligned} E_+ &= 107201 + (\cos(\pi))^2 127221 \\ &= 107201 + (0)^2 127221 \\ &= \underline{\underline{107201}} \end{aligned}$$

$$\begin{aligned} E_- &= (\sin(\pi))^2 127221 \\ &= (0)^2 127221 \\ &= 127221 \\ &= \underline{\underline{127221}} \end{aligned}$$

System Initial mixed state is

$$|u_1\rangle = |0\rangle, P_1 = \frac{1}{4}$$

$$|u_2\rangle = |1\rangle, P_2 = \frac{1}{2}$$

$$|u_3\rangle = |2\rangle, P_3 = \frac{2}{3}$$

$$\begin{aligned} S_f &= \sum_{i=1}^3 P_i |u_i\rangle \langle u_i| = P_1 |u_1\rangle \langle u_1| + P_2 |u_2\rangle \langle u_2| + P_3 |u_3\rangle \langle u_3| \\ &= \frac{1}{4} (107201) + \frac{1}{2} (127221) + \frac{2}{3} (142414) \end{aligned}$$

8)

$$\begin{aligned}
 S_5 &= \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{12} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{2}{3} \cdot \frac{1}{2} \begin{pmatrix} 1 & -\frac{1}{3} \\ \frac{1}{3} & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{12} \end{pmatrix} + \begin{pmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{4} + \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{12} + \frac{1}{3} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{7}{12} & -\frac{1}{3} \\ \frac{1}{3} & \frac{5}{12} \end{pmatrix} \\
 &\quad \underline{\underline{\quad \quad \quad}}
 \end{aligned}$$

Probability distribution

$$P_+ = \text{Tr}[S_5 E_+]$$

$$\begin{aligned}
 P_+ = \text{Tr}[S_5 E_+] &= \text{Tr} \left[\begin{pmatrix} \frac{7}{12} & -\frac{1}{3} \\ \frac{1}{3} & \frac{5}{12} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right] \\
 &= \text{Tr} \begin{pmatrix} \frac{7}{12} & 0 \\ \frac{1}{3} & 0 \end{pmatrix} = \frac{7}{12} + 0 = \frac{7}{12} \\
 &= \underline{\underline{\frac{7}{12}}}
 \end{aligned}$$

$$\begin{aligned}
 P_- = \text{Tr}[S_5 E_-] &= \text{Tr} \left[\begin{pmatrix} \frac{7}{12} & -\frac{1}{3} \\ \frac{1}{3} & \frac{5}{12} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right] \\
 &= \text{Tr} \begin{pmatrix} 0 & -\frac{1}{3} \\ 0 & \frac{5}{12} \end{pmatrix} = \frac{5}{12} + 0 = \frac{5}{12} \\
 &= \underline{\underline{\frac{5}{12}}}
 \end{aligned}$$

$$\sum p_i = 2$$

$$\sum p_i = p_+ + p_- = \frac{7}{12} + \frac{5}{12} = \frac{12}{12} = 1$$

$$\underline{\underline{\sum p_i = 1}}$$

, The outcome probability distribution

$$p(+)=\frac{7}{12}$$

$$p(-)=\frac{5}{12}\underline{\underline{\quad}}$$